The Tangent Line Triangle of $f(x) = \frac{1}{x}$ A Calculus vs. Geometric Approach

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1 Definition

Calculate the area of triangle ABO, formed from the axis intersections of a tangent line to the function $f(x) = \frac{1}{x}$, at the point $(g, f(g))$. (Fig 1.)

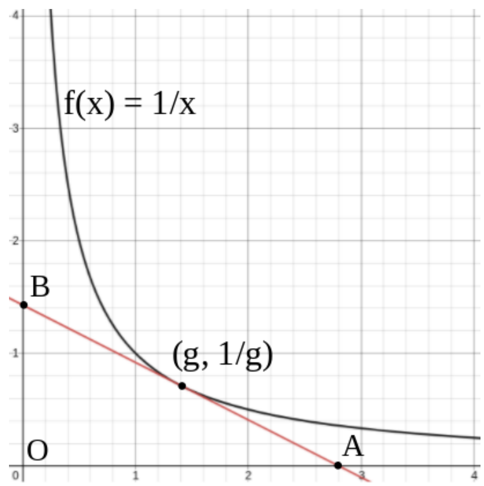


Figure 1: Graph of $f(x) = \frac{1}{x}$ and its tangent line at the point $(g, f(g))$.

2 The Calculus-Based Solution

The general equation for the tangent line to a function, $f(x)$, at the point of tangency $(g, f(g))$ is shown below. For proof see Appendix 1.

$$y = f'(g)(x - g) + f(g)$$

For $\frac{1}{x}$, the tangent-line function (at the x-value of g) is;

$$y = -\frac{1}{g^2}(x - g) + \frac{1}{g}$$

$$y = \frac{-1}{g^2}x + \frac{2}{g}$$

The area of triangle ABO can be evaluated by the integration of this tangent line equation, from $x = 0$ to point A. Hence, the x-intercept, A, must be located.

$$0 = \frac{-1}{g^2}x + \frac{2}{g}$$

$$\frac{\frac{-2}{g}}{\frac{-1}{g^2}} = x$$

$$x = 2g$$

Thus, the integral must be evaluated from $x = 0$ to $x = 2g$.

The anti-derivative is computed ($y = \frac{-1}{2g^2}x^2 + \frac{2}{g}x$), the Fundamental Theorem of Calculus is applied, and the function may now be integrated.

$$\int_0^{2g} \left(\frac{-1}{g^2}x + \frac{2}{g}\right) dx = \left(\frac{-1}{2g^2}(2g)^2 + \frac{2}{g}(2g)\right) - \left(\frac{-1}{2g^2}0^2 + \frac{2}{g}0\right)$$

$$\int_0^{2g} \left(\frac{-1}{g^2}x + \frac{2}{g}\right) dx = \left(\frac{-1}{2g^2}4g^2 + 4\right)$$

$$\int_0^{2g} \left(\frac{-1}{g^2}x + \frac{2}{g}\right) dx = (-2 + 4)$$

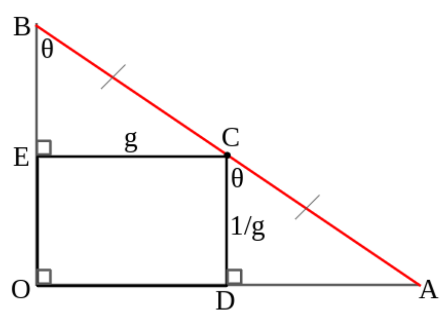
$$= 2$$

Interestingly, the tangent line triangle, ABO, has a constant area of 2, regardless of the point of tangency.

3 The Geometric Technique

Although integration can be employed to calculate area, the traditional geometric method, $\frac{1}{2} \cdot base \cdot height$, can also be used.

As shown in the image below, the tangent line triangle problem can be expressed as a geometric figure, where the point of tangency is C, and sides OA and OB represent the x and y axes.



Both triangles EBC and DCA are similar to triangle OBA (AAA Similarity), and as such are similar to each other.

Knowing that segment BC is equal to CA (See Appendix 2), triangles EBC and DCA are congruent (AAS Congruence), meaning $OB = \frac{1}{g} + \frac{1}{g}$ and $OA = g + g$. The area is calculated:

$$\frac{OB \cdot OA}{2} = \frac{\frac{2}{g} \cdot 2g}{2} = \frac{4}{2} = 2$$

Here too, the geometric method also shows the constant area property of the tangent line triangle.

4 The General Tangent Line Case

To calculate the area of the tangent line triangle for any function, $f(x)$, the 'geometric' method, $\frac{1}{2} \cdot base \cdot height$, will be used.

4.1

The x (the 'base' of the triangle) and y (the 'height' of the triangle) intercepts of the general tangent line equation, $y = f'(g)(x-g) + f(g)$ (See Appendix 1), are calculated as follows:

x-intercept:

$$0 = f'(g)(x-g) + f(g)$$

$$x = \frac{-f(g)}{f'(g)} + g$$

y-intercept:

$$y = f'(g)(0-g) + f(g)$$

$$y = -g \cdot f'(g) + f(g)$$

The Tangent Line Triangle's Area:

$$\frac{\left(-g \cdot f'(g) + f(g)\right)\left(\frac{-f(g)}{f'(g)} + g\right)}{2}$$

$$Area = \frac{2g \cdot f(g) - g^2 \cdot f'(g) - \frac{f(g)^2}{f'(g)}}{2}$$

For illustrative purposes, we will apply the general area equation to the related function $f(x) = \frac{1}{kx^n}$, whose derivative, $f'(x)$, is $\frac{-1}{k^2x^{2n}} \cdot nkx^{(n-1)}$ or $\frac{-nx^{(-n-1)}}{k}$.

$$Area = \frac{2g \cdot f(g) - g^2 \cdot f'(g) - \frac{f(g)^2}{f'(g)}}{2} = \frac{2g \cdot \frac{1}{kg^n} - g^2 \cdot \frac{-ng^{(-n-1)}}{k} - \frac{\frac{1}{k^2g^{2n}}}{\frac{-ng^{(-n-1)}}{k}}}{2}$$

$$\frac{\frac{2g^{(1-n)}}{k} + \frac{ng^{(-n+1)}}{k} + \frac{1}{nkg^{(n-1)}}}{2}$$

$$\frac{2g^{(1-n)} + ng^{(1-n)} + \frac{1}{ng^{(n-1)}}}{2k}$$

$$Area = \frac{g^{(1-n)}\left(2 + n + \frac{1}{n}\right)}{2k}$$

It can be seen that the x-value of the point of tangency, g, appears in the tangent-line-triangle-area equation. This is true for all n-values other than 1. Limiting our focus to $f(x) = \frac{1}{kx}$, the tangent-line-triangle has a constant area of:

$$\frac{g^{(0)}(2+1+1)}{2k}$$

$$Area = \frac{2}{k}$$

For a k-value of 1, as in $f(x) = \frac{1}{x}$, the area is correspondingly $\frac{2}{1}$, the value previously found.

4.2

Note that the midpoint C congruency argument of Section 3 is also true for any value of k within the tangent line equation for $f(x) = \frac{1}{kx}$, $y = \left(\frac{-1}{kg^2}\right)(x-g) + \frac{1}{kg}$.

x-intercept:

$$x = \frac{-\frac{1}{kg}}{\left(\frac{-k}{k^2g^2}\right)} + g = 2g$$

y-intercept:

$$y = \left(\frac{-1}{kg^2}\right)(-g) + \frac{g}{kg^2} = \frac{2g}{kg^2} = \frac{2}{kg}$$

Where the midpoint is then:

$$\left(\frac{2g+0}{2}, \frac{\frac{2}{kg}+0}{2}\right)$$

$$\left(g, \frac{1}{kg}\right)$$

The coordinate of midpoint C of line AB remains the point of tangency.
Applying the area rule from Section 3: $OB = \frac{1}{kg} + \frac{1}{kg}$ and $OA = g + g$

$$Area = \frac{\frac{2}{kg} \cdot 2g}{2} = \frac{2}{k}$$

The identical expression found in Section 4.1.

5 Appendix

1. The equation of a tangent line to a function $f(x)$ in the form of $y = mx + c$, has a slope, m, equal to the derivative of the function, $f'(x)$, evaluated at the x-coordinate of the point of tangency, g.

$$y = f'(g)\,x + c$$

At this point of tangency, $(g, f(g))$, the equation must hold true, and as such the c-value can be obtained by rearrangement.

$$f(g) = f'(g) \cdot g + c$$
$$c = f(g) - f'(g) \cdot g$$

Where substitution into the original formula yields;

$$y = f'(g)\,x + f(g) - f'(g) \cdot g$$

$$y = f'(g)(x - g) + f(g)$$

2. The intercepts of the $\frac{1}{x}$ tangent-line equation $y = \frac{-x}{g^2} + \frac{2}{g}$ (See Section 2) are $\left(0, \frac{2}{g}\right)$ and $(2g, 0)$, thus their midpoint (the average of their x and y components) is:

$$\left(\frac{2g+0}{2}, \frac{0+\frac{2}{g}}{2}\right)$$

$$\left(g, \frac{1}{g}\right)$$

Therefore, the point of tangency C is equidistant from the x and y intercepts of the tangent line AB.

3. Observe yet another constant-area property of the $f(x) = \frac{1}{kx}$ function: any rectangle formed between the origin and point $\left(g, \frac{1}{kg}\right)$, rectangle ODCE, also has a constant area of $g \cdot \frac{1}{kg}$, or $\frac{1}{k}$ (no matter the g-value selected).